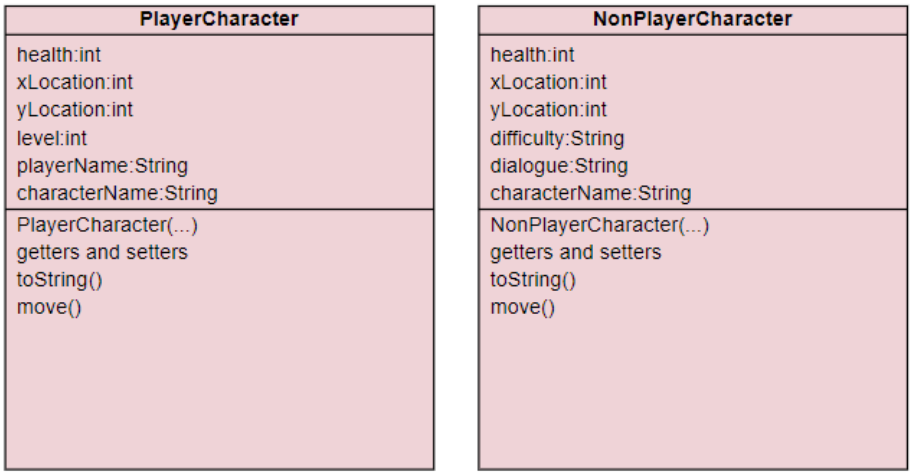
Programming Fundamentals II: Lab 07

Due Date: Tuesday October 29, 2024 at Midnight

Part 1. Video Game Entities (10 Marks)

Suppose we have the following classes implemented a system for a simple video game:



a. **(4 Marks)** Create a generalized class for PlayerCharacter and NonPlayerCharacter called Entity. Your generalized class should remove any redundant fields and methods from the PlayerCharacter and NonPlayerCharacter class. Your generalized class should not be able to be instantiated. Ensure all fields in the general class have protected access. Include only a full-arg constructor for your class.

b. **(2 Marks)** The move() method should accept two ints as input, which represent the x and y location to move to. The move method should then set the x and y values of the entity that is moving.

c. **(4 Marks)** Write the code for your PlayerCharacter and NonPlayerCharacter classes. Include

   a. The toString method for PlayerCharacter should return the character name followed by the player name in parentheses
   b. The toString method for NonPlayerCharacter should return the character name followed by their dialogue in quotes

Sample run

```
public static void main(String[] args){
    PlayerCharacter pc = new PlayerCharacter(2, 10, 0, 4, “John”, “Link”);
    NonPlayerCharacter npc =
      new NonPlayerCharacter (5, 40, 0, “Hard”, "It’s a secret to everyone",
      "Moblin");

    System.out.println(pc);
    System.out.println(npc);
}
```

Will print out

```
Link (John)
Moblin: “It’s a secret to everyone”
```

Add the **date and your name and student number** to the top of your java files. Please submit your java files to Nexus before the end of the day on the due date. **Late submissions will be not be accepted.**